

choptuik_ac_bc · theorems12 · github.com/wild8highlander/choptuik_ac_bc

The dd-Closure Theorem: the uniqueness of the prolongation and the Jordan-2 nilpotent

rank $U_k = 9$ exactly · $B = [[0, -350/61], [0, 0]]$, $B^2 = 0$ · $\delta_{\text{mono}} = 0$ linearly

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Series:	Monograph No. 05 of the “Twelve Theorems” cycle
Machine:	sympy_dd_closure.py · SymPy, exact in $\mathbb{Q}(\sqrt{3})$
Verdict:	einstein_direct/results/dd_closure.json
Status:	PROVED EXACTLY (baseline) + numerical (brick)

Abstract

This monograph records the theorem of the independent dd-law of evolution of the truncated tower: the law of second derivatives is derived from the identities, not chosen. On the purified forms the y -prolongation of the residues coincides with the τ -prolongation (the weight rule), so the closure of the prolongation is the preservation of the constraints; the rank of the prolongation matrix is $\text{rank } U_k = 9$ exactly over $\mathbb{Q}(\sqrt{3})$ — the closure exists and is unique, the hidden conditions have rank 5, the index reduction lowers the index from three to two, and the admissible manifold M_2 is two-dimensional.

The evolution on M_2 is closed by the matrix $B = \begin{bmatrix} 0 & -350/61 \\ 0 & 0 \end{bmatrix}$ with $B^2 = 0$ exactly — a nilpotent Jordan-2 block: statics plus a linear secular drift, without exponents or oscillations; $\delta_{\text{mono}} = 0$ in the closed linear dynamics, and the residual $\sim 2.9\%$ to $\pi/30$ must be finite-amplitude. The alternative closures (adiabatics, chain-pinned flows) are shown degenerate by the machine. The rank tables of both points, the corollaries, the honest caveats and the reproduction protocol are included.

Keywords: dd-closure, prolongation, hidden conditions, Jordan-2 nilpotent, $\delta_{\text{mono}} = 0$, SymPy, machine proof.

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1. Introduction

The fifth monograph records the theorem of the independent dd-law of evolution: the closure of the prolongation is unique, the admissible manifold is two-dimensional, and the closed linear dynamics on it is a nilpotent Jordan-2 block with the matrix $B = \begin{bmatrix} 0, & -350/61 \\ 0, & 0 \end{bmatrix}$. The result was delivered by campaign v14 (machine `sympy_dd_closure.py`, verdict `dd_closure.json`, session 18b, LEDGER §25) and answers the central methodological question of the programme: the dd-slots were introduced with coefficient 1 “by the ddQ precedent” — can the evolution law of the second flows be justified from the identities rather than chosen or fitted?

The answer is yes, and uniquely so. On the purified forms the derivative with respect to log-time coincides with the τ -derivative (the weight rule, monograph No. 4), so the prolongation of the residues is the preservation of the constraints; the prolongation system contains nine unknown evolutionary derivatives and is solved uniquely (rank $U_k = 9$ exactly over $Q(\sqrt{3})$). The consistency of the prolongation produces five hidden conditions, the index reduction lowers the index of the system from three to two, and on the two-dimensional admissible manifold M_2 the evolution is closed by a matrix B with $B^2 = 0$ exactly: the spectrum is $\{0, 0\}$, no exponents, no oscillations. The consequence — $\delta_{\text{mono}} = 0$ in the closed linear dynamics — separates the fate of the tower from the finite-amplitude mechanisms.

2. Notation and Setting

The state of the system is a vector of dimension 20: nine amplitudes $a[9]$, eight of their first derivatives $v[8]$ and three second derivatives $dd[3]$. The constraints $J1s \cdot x = 0$ are the collection of the tower equations and chain links; the prolongation $d/d\tau(J1s \cdot x) = 0$ with nine unknowns ($M5h'$, $ddT0$, $ddD0$, ddQ , $ddR1$, $ddR5$, $dddW2$, $dddR3$, $dddP4$) gives the linear system $U_k \cdot u = -Known \cdot (v, dd)$. The consistency of this system with respect to the right-hand side — the hidden conditions $Op \cdot (v, dd) = 0$; the manifold $M_2 = \ker[J1s; Op']$ is the set of states consistent with the constraints and their prolongation.

The setting: to prove that (a) the y -prolongation of the residues is identically equal to the τ -prolongation (the law is not chosen); (b) the rank of U_k is nine — the closure exists and is unique; (c) the hidden conditions have rank five, M_2 is two-dimensional; (d) the evolution on M_2 is given by a unique matrix B , to write it out exactly and determine its Jordan structure; (e) to show that the alternative closures are degenerate.

3. Statement of the Theorem

Theorem 5 (the dd-closure and the Jordan-2 nilpotent). For the corrected prolonged system on the static chain:

(i) the justification of the law is exact: on the purified forms $d/dy = (1/s) \cdot d/d\tau$ (the weight rule), the y -prolongation of the residues coincides with the τ -prolongation, so the closure is derived from the identities (Bianchi), not chosen;

(ii) at the baseline point ($\kappa = 2$, $\tau^* = 27/4$) the rank of the prolongation matrix is $\text{rank } U_k = 9$ exactly (the field $Q(\sqrt{3})$): the closure $U_k \cdot u = -\text{Known} \cdot (v, dd)$ exists and is unique; the hidden conditions have $\text{rank } O_p = 5$; $\text{rank}[J_{1s}; O_p'] = 17$; the index reduction passes two levels ($3 \rightarrow 2$);

(iii) the admissible manifold M_2 is two-dimensional ($\dim M_2 = 2$), the closure lies in it with residuals 0.0 exactly, and the evolution on M_2 is given by the matrix

$$B = [[0, -350/61], [0, 0]], \quad B^2 = 0 \quad (5.1)$$

— a nilpotent Jordan-2 block: statics plus a linear secular drift, without exponents or oscillations; on the one-brick point the same structure is reproduced numerically ($\|B^2\| = 3.9 \times 10^{-13}$, eigenvalues $\pm 2.5 \times 10^{-7}$ — the numerical zero of the defective block);

(iv) the alternatives are degenerate: the adiabatic closure $dd = 0$ is compatible only with the frozen set (the common branch factor $4T_0h^2 - 27$, R_{1h} free — a one-dimensional manifold); the flows pinned to the chain give a single closed point — the frozen one; the prolongation closure is the only justified one.

4. Sketch of the Machine Proof

4.1. The justification of the law

Layer B1 of the machine establishes the chain of justification: the residues are Bianchi-reduced Einstein equations — identities in y ; the y -prolongation of an identity remains an identity; on the purified forms $d/dy = (1/s)(d/d\tau + \text{weight})$, where the weight term factorizes by the weight rule (monograph No. 4); hence the τ -prolongation of the residues is equivalent to the preservation of the constraints, and the evolution law dd is the unique solution of the preservation, not a choice. `pure_form_exact = true` and `weight_rule_exact = true` in the verdict `pin` both links.

4.2. The uniqueness of the closure

Layer B2 at the baseline assembles the prolongation system symbolically and lambdaifies it in exact arithmetic over $Q(\sqrt{3})$: rank $U_k = 9$ with a null tail of singular values at machine zero; the hidden conditions have rank 5 (tail 10^{-14} – 10^{-16} in the numerical control of the brick point); $\text{rank}[J1s; Op'] = 17$ in the 20-dimensional state, which leaves $\dim M2 = 2$. The index analysis: the level-3 violation has rank 1, level-4 rank 0; two reduction iterations bring the system to index 2. The closure solved on $M2$ yields vectors with residuals $\text{closure_in_M} = (0.0, 0.0)$ exactly.

4.3. The nilpotent and the alternatives

The matrix B is obtained in closed form and verified exactly: $B^2 = 0$, rank $B = 1$, eigenvalues $\{0\}$ — the classic defective Jordan-2 block. Layer B4 enumerates the alternatives: (C0) the adiabatics $dd = 0$ is compatible only on the frozen set, where the branches $UV_{\xi 3}$ and $Mdef_{\xi 5}$ share the common factor $4T0h^2 - 27$ (see (3.1) of monograph No. 3); (C-chain) the flows with $d(\text{chain})/d\tau = 0$ close only at the frozen point (session 13, honest quote); (C-prolongation) — the only justified alternative, and it is our theorem.

$$UV_{\xi 3} = -(16/27)R1h^3T0h^2(4T0h^2 - 27) = -(64/27)R1h^3T0h^2(T0h^2 - 27/4) \quad (5.2)$$

The factorization (5.2) — the branches of the frozen set: outside the factor $4T0h^2 - 27$ the adiabatics is incompatible with the constraints, so “freezing” is not a dynamics, and the only live dynamics is the prolongation one.

5. Numerical Results

Table 5.1. The rank structure of the dd-closure (baseline $\kappa = 2$ exact; brick — numerically)

Quantity	$\kappa = 2$ (EXACT)	$\kappa = 2 - \pi^2/98$ (float64)
τ^*	6.75000000000000000000	7.1079202069226305031
rank Uk	9	9 (tail $2.9 \times 10^{-2} \dots$)
rank Op (hidden)	5	5 (tail 5.5×10^{-16})
rank[J1s; Op']	17	17
dim M2	2	2
closure residual in M2	0.0 / 0.0	1.9×10^{-13} / 5.9×10^{-14}
$\ B^2\ $	0 exactly	3.9×10^{-13}
spec B	{0, 0}	$\pm 2.5 \times 10^{-7}$ (defective)

The numerical column is an honest control: the structure reproduces in float64 with clean singular-value gaps; the eigenvalues $\pm 2.5 \times 10^{-7}$ are the typical numerical trace of the defective zero, not a physical pair of modes. Additional marches (march_delta_mono) measure the secular growth by two independent realizations of the closure: the log-log slope $\rightarrow 1$ (the linear secularity of Jordan-2), $\delta_mono < 2.8 \times 10^{-14}$ per echo.

6. Corollaries

Corollary 5.1 ($\delta_mono = 0$ linearly). The closed linear dynamics on M2 is a Jordan-2 nilpotent: per echo only a secular drift is possible, there is no monodromy phase shift. Any residual “baryon asymmetry” ($\sim 2.9\%$ to $\pi/30$) must be finite-amplitude — the v14–v15 programme is built exactly on this separation.

Corollary 5.2 (the law is derived). Layers B1–B2 close the methodological gap of v13: the dd-law is no longer a “precedent” but a theorem. Together with the weight rule

(monograph No. 4) this makes the whole de-adiabatized system derivable from the identities, without a single fitted constant.

Corollary 5.3 (a stub for the march theorem). The two-dimensionality of M2 and the exact form of B are the inputs of the nonlinear march of v15: the linear picture is known exactly, so any deviation at finite amplitude is attributed to the nonlinearity, not to the linearization (monograph No. 6).

7. Honest Caveats

The exact column of Table 5.1 refers to the baseline; on the brick the structure is established numerically (float64) with clean gaps, the symbolic version for $\kappa = 2 - \pi^2/98$ was not built — the field $\mathbb{Q}(\pi)$ is substantially more expensive, and the two-point correlation (Theorem 2) is deemed sufficient control. The tower truncation: the theorem describes the dynamics of the truncated system with the derived dd-law; a tower enhancement changes the dimensions (9 prolongation unknowns — a consequence of the truncation) and requires a rebuild. Puiseux branches at the soft pole $45/8$ may change the chain representation; the closure is built in the T1c source form. Finally, δ_{mono} is measured by the march protocol down to 10^{-14} per echo; below that is the limit of the float64 representation, not physics.

8. Reproducibility

Table 5.2. Reproduction protocol for Theorem 5

Step	Command	Result
1	<code>python3 sympy_dd_closure.py</code>	results/dd_closure.json (layers B1–B4, the B matrix)
2	<code>python3 sympy_third_order.py</code>	results/third_order_tables.json (the weight rule)
3	<code>python3 march_delta_mono.py</code>	results/march_delta_mono.json (δ_{mono} by two marches)
4	<code>python3 -m pytest tests/ -q</code>	58 passed (the closure tests included)

The running time is 35.6 s (runtime_s) for step 1 on an ordinary laptop; the machines are single-threaded. The data correspond to commit 5b256cd; an autonomous copy lives in `einstein_direct/open9/machines/`.

References

- [1] Repository `wild8highlander/choptuik_ac_bc`: `einstein_direct/sympy_dd_closure.py`; `einstein_direct/results/dd_closure.json` (campaign v14, session 18b, LEDGER §25).
- [2] Ibid.: `einstein_direct/sympy_third_order.py`; `einstein_direct/results/third_order_tables.json` (the weight rule).
- [3] Ibid.: `einstein_direct/march_delta_mono.py`; `einstein_direct/results/march_delta_mono.json` (two independent marches).
- [4] Ibid.: `einstein_direct/dae_nonlinear_core.py`; `einstein_direct/results/march_dae_nonlinear.json` (the nonlinear continuation, v15).
- [5] Monographs No. 1, No. 4, No. 6 of the theorems12 cycle.